

# NVSHU: Virtual Reality Design and Narrative Popularization for Intangible Cultural Heritage Characters

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## ABSTRACT

This project reconstructs and reproduces the ecological context of “NVSHU” culture in an immersive (Virtual Reality) VR experience. We explore interactive approaches of communication for this unique intangible heritage context. “NVSHU” amplifies the effectiveness of intangible cultural heritage (ICH) preservation through narrative interaction and virtual character popularization techniques, aiming to enhance audiences’ experiential learning, and augment their sense of presence and immersion.

## CCS CONCEPTS

• **Computing methodologies** → **Virtual reality**; • **Human-centered computing** → **Interaction design theory, concepts and paradigms**.

## KEYWORDS

Virtual reality, Narrative interaction, Character learning

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## 1 INTRODUCTION

NVSHU, the sole gender-specific script worldwide—specifically female, originating from Jiangyong, China, served as a communication tool employed by women who had undergone foot-binding (“three-inch golden lotus”). Nonetheless, NVSHU, in its original ecological framework, confronts a crisis of deconstruction within the realm of public discourse. Through enhancing language learning technology and virtual reality gaming, as demonstrated by [Cheng et al. 2017], this research further delves into the potentialities of virtual reality design and narrative popularization for the preservation of intangible cultural heritage text, focusing on narrative interactivity and VR character popularization.

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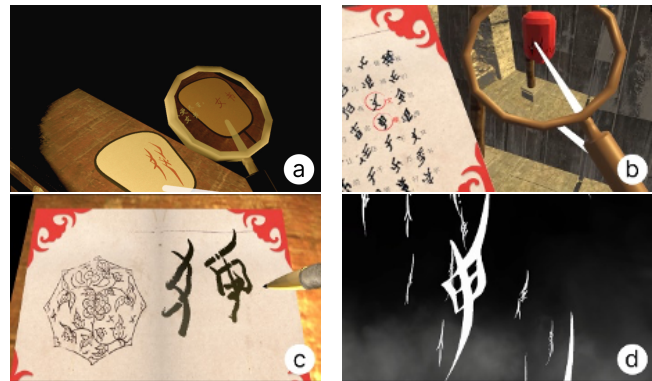


Figure 1: (a) Narrative Interaction, VR Character Popularization: (b) Learning: Translation and Songs. (c) Application: Free Writing. (d) Experience: Immersive Character Space.

## 2 NARRATIVE INTERACTION

To engender an elevated sense of presence, the VR environment should be integrally associated with a contextualized and interactive digital narrative [Irshad and Perkiş 2020]. Within “NVSHU”, both interactive behaviors and items serve dual purposes, namely, storytelling and science popularization. Leveraging the concept of stealth and freedom, we have designed characters that vary in visibility (see Figure 1a). NVSHU characters appear as decorative patterns, requiring users to employ a magnifying glass for identification.

In Scene 1, the experienter picks up the paper-cut clipped in the “Book of Three Dynasties” and enters the memories of her old times. In Scene 2, the experienter need to complete the first three pages written by the owner’s female relatives and friends and listen to their songs. The last few pages of the book are often the autobiography of the owner, in Scene 3, the experienter can pick up a pen and write freely on the page.

## 3 VR CHARACTER POPULARIZATION

Compared to traditional methods, language learning within VR offers a more engaging experience due to the implementation of gamification and the consequent less academic ambiance [Garcia et al. 2019]. In response to challenges related to fonts and dialect pronunciation within NVSHU, we have devised a gradual learning process to promote NVSHU culture.

### 3.1 Learning: Translations and Songs

"Reading and singing with papers and fans" is the principal methods for women to communicate and learn NVSHU. In Scene 1, we designed a form of comparison between NVSHU and Chinese, which is convenient for the experimenter to have a preliminary understanding of unfamiliar characters. In Scene 2, the experimenter triggers the interactive completion of the content in the book and learning through gamified interactive settings (see Figure 1b).

### 3.2 Application: Free Writing

Post an engaging exploration, users are encouraged to freely write in Scene 3 (see Figure 1c). Observing, listening, and writing the text can fortify the impressions of the users and foster unique creativity.

### 3.3 Feeling: Immersive Character Space

In Scene 4, the text particles and chorus singing lead the experimenter to immerse themselves in the beautiful world of NVSHU (see Figure 1d). They can feel the charm of the character and the feminism contained in NVSHU from the scene.

## 4 SUMMARY

This paper consolidates various modes of expressing intangible cultural heritage characters represented by NVSHU within VR. Through narrative interaction and VR character popularization, we display NVSHU culture in its original ecological communication environment, offering users enhanced presence and memorable touch-points for text popularization. This work serves as a preliminary design reference for future applications of VR in the preservation of intangible cultural heritage characters.

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